GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 6

COURSE: SOFTWARE ENGINEERING (ETCS-601)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETCS-601

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Define Software Engineering and state IEEE definition of software.

Q1. (b) Differentiate between Waterfall model and Spiral risk-driven model.

Q1. (c) What is SRS (Software Requirements Specification)? State characteristics of a good SRS.

Q1. (d) What is difference between functional and non-functional requirements?

Q1. (e) Define cohesion and coupling. Why is high cohesion and low coupling desirable?

Q1. (f) Differentiate between Verification and Validation in software engineering.

Q1. (g) What is Cyclomatic Complexity? State McCabe formula v(G) = e - n + 2p.

Q1. (h) What is regression testing and smoke testing?

Q1. (i) Define COCOMO model: Basic, Intermediate, and Detailed.

Q1. (j) What is Agile Manifesto? List 4 core values of Agile methodology.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Compare Classical Waterfall, Prototyping, RAD, and Agile Scrum models with lifecycle phases.

Q2. (b) Develop complete SRS document for an Online Food Delivery Platform according to IEEE 830 standard.

Q2. (c) For given control flow graph with 14 edges and 10 nodes, calculate cyclomatic complexity and identify independent paths.

Q2. (d) Explain Black-box testing techniques: Boundary Value Analysis (BVA), Equivalence Class Partitioning (ECP), and Cause-Effect

Q2. (e) Explain Object-Oriented Design principles (SOLID): Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregati

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Draw UML Use Case, Class, and Sequence diagrams for an ATM Banking System. [10 Marks]

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(b) Explain architectural patterns: Layered, Client-Server, Microservices, and Event-Driven architecture. [10 Marks]

Q4. (a) Explain software metrics: Function Point (FP) analysis and calculate FP for given transaction inputs. [10 Marks]

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(b) Explain COCOMO-II estimation model and calculate effort and duration for 50 KLOC project. [10 Marks]

Q5. (a) Explain White-box testing: Statement coverage, Branch coverage, Condition coverage, and Path coverage. [10 Marks]

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(b) Explain test automation frameworks: Unit testing, Integration testing (Top-down vs Bottom-up), System testing. [10 Marks]

Q6. (a) Explain Software Quality Assurance (SQA) and CMMI (Capability Maturity Model Integration) 5 levels. [10 Marks]

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(b) Explain ISO 9001 and Six Sigma quality standards in software development processes. [10 Marks]

Q7. (a) Explain Scrum framework: Product Backlog, Sprint Backlog, Daily Standup, Sprint Review, and Retrospective. [1

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(b) Explain DevOps lifecycle, Continuous Integration (CI), and Continuous Deployment (CD) pipelines. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****